

Power Query Data Cleaning & Transformation

Learning Objective:

The objective of this phase is to clean and transform the retail and e-commerce sales dataset using Microsoft Power Query. The raw Orders and Returns datasets were prepared for analysis by correcting data quality issues, assigning appropriate data types, removing duplicate and missing records, validating business rules, and ensuring consistency before loading the data into Power BI.

Power Query Activities:

1. Imported Orders and Returns tables into Power Query.
2. Renamed queries for better readability.
3. Assigned correct data types to all columns.
4. Converted Order Date and Ship Date using English (India) locale.
5. Applied Trim and Clean to text columns.
6. Standardized Category values.
7. Validated Brand values and retained official brand names.
8. Corrected invalid Discount values.
9. Converted Discount column to Decimal Number.
10. Corrected negative Quantity values.
11. Removed duplicate records from the Returns dataset.
12. Removed rows containing NULL values in mandatory columns.
13. Validated business rules after cleaning.
14. Loaded the cleaned data back into Excel.

Transformation Log:

Step	Transformation	Reason	Status
1	Imported Orders table into Power Query	Load raw data for cleaning	Completed
2	Imported Returns table into Power Query	Load return records	Completed
3	Renamed Orders and Returns queries	Improve readability	Completed
4	Changed data types for all columns	Ensure correct data types	Completed
5	Converted Order Date to Date using English (India) locale	Standardize date format	Completed

Step	Transformation	Reason	Status
6	Converted Ship Date to Date using English (India) locale	Standardize date format	Completed
7	Applied Trim to text columns	Remove leading and trailing spaces	Completed
8	Applied Clean to text columns	Remove hidden and non-printable characters	Completed
9	Standardized Category values	Maintain consistent category names	Completed
10	Validated Brand names	Retained official brand names such as HP, IKEA and JK	Completed
11	Replaced invalid Discount value "abc" with NULL	Prepare for numeric conversion	Completed
12	Corrected invalid Discount values (-5, 55, 65, 120)	Ensure valid discount values	Completed
13	Converted Discount column to Decimal Number	Correct numeric data type	Completed
14	Corrected Quantity values from -3 to 3	Follow business rules	Completed
15	Removed duplicate rows from Returns table	Remove duplicate records	Completed
16	Removed rows containing NULL values in Customer Name	Improve data quality	Completed
17	Removed rows containing NULL values in Discount	Remove incomplete records	Completed
18	Validated Customer Name, State and Category NULL rows	Same records removed through filtering	Completed
19	Verified Sales, Quantity and Discount consistency	Business validation	Completed
20	Loaded cleaned data back into Excel	Prepare for Power BI analysis	Completed